

# ActionShap: Supplementary Numerical Audit

Schema-v2 release; source commit **b778f55**; result  
archive SHA-256

**ac4c7fb1993458b6b41054974ebff215710e7a8b5894c7aa6af828e94b2a5b0f**

Mouad Louhichi<sup>1\*</sup>, Redwane Nesmaoui<sup>1</sup> and Mohamed Lazaar<sup>1</sup>

<sup>1\*</sup>National Higher School of Computer Science and Systems Analysis  
(ENSIAS), Mohammed V University in Rabat, Rabat, Morocco.

\*Corresponding author(s). E-mail(s): [mouad\\_louhichi@um5.ac.ma](mailto:mouad_louhichi@um5.ac.ma);

Schema-v2 audit; source commit and archive SHA-256 are recorded in **RELEASE.METADATA.md**.

This supplementary document contains the expanded numerical audit behind the compact manuscript tables. It is generated from the schema-v2 release and is not required for compiling the main manuscript. The CSV/JSON release remains the authoritative machine-readable source.

## S1 Alignment matrix (Table S1)

**Table S1:** Complete singleton alignment matrix for all predeclared ItemKNN conditions. Means are distinct-user values; confidence intervals and missing-user counts are retained in the CSV release.

| Condition                  | Method                           | Deletion | Bounded | Gap     | $n$ |
|----------------------------|----------------------------------|----------|---------|---------|-----|
| Amazon full cata-<br>logue | Greedy<br>sequential<br>deletion | 0.5731   | 0.4494  | -0.1236 | 250 |
| Amazon full cata-<br>logue | LIME                             | 0.9215   | 0.7215  | -0.2000 | 250 |

Continued on next page

Table S1 – continued from previous page

| Condition                | Method                     | Deletion | Bounded | Gap     | $n$ |
|--------------------------|----------------------------|----------|---------|---------|-----|
| Amazon full catalogue    | Leave-one-out              | 1.0000   | 0.7895  | -0.2105 | 250 |
| Amazon full catalogue    | Monte Carlo Shapley        | -0.2875  | -0.1186 | 0.1689  | 250 |
| Amazon full catalogue    | Random control             | -0.0012  | -0.0041 | -0.0028 | 250 |
| Amazon sampled           | Greedy sequential deletion | 0.1911   | 0.1310  | -0.0600 | 993 |
| Amazon sampled           | LIME                       | 0.9304   | 0.8271  | -0.1032 | 993 |
| Amazon sampled           | Leave-one-out              | 1.0000   | 0.8511  | -0.1489 | 993 |
| Amazon sampled           | Monte Carlo Shapley        | 0.2856   | 0.4141  | 0.1285  | 993 |
| Amazon sampled           | Random control             | -0.0080  | -0.0103 | -0.0023 | 993 |
| MovieLens 100 candidates | Greedy sequential deletion | 0.2632   | 0.2605  | -0.0027 | 250 |
| MovieLens 100 candidates | LIME                       | 0.9604   | 0.9469  | -0.0136 | 250 |
| MovieLens 100 candidates | Leave-one-out              | 1.0000   | 0.9824  | -0.0176 | 250 |
| MovieLens 100 candidates | Monte Carlo Shapley        | 0.8086   | 0.8209  | 0.0124  | 250 |
| MovieLens 100 candidates | Random control             | -0.0044  | -0.0029 | 0.0015  | 250 |
| MovieLens 500 candidates | Greedy sequential deletion | 0.3733   | 0.3782  | 0.0049  | 250 |
| MovieLens 500 candidates | LIME                       | 0.9445   | 0.9219  | -0.0226 | 250 |
| MovieLens 500 candidates | Leave-one-out              | 1.0000   | 0.9710  | -0.0290 | 250 |
| MovieLens 500 candidates | Monte Carlo Shapley        | 0.6918   | 0.7239  | 0.0321  | 250 |
| MovieLens 500 candidates | Random control             | 0.0018   | 0.0040  | 0.0022  | 250 |
| MovieLens full catalogue | Greedy sequential deletion | 0.3845   | 0.3819  | -0.0026 | 250 |
| MovieLens full catalogue | LIME                       | 0.9256   | 0.8924  | -0.0332 | 250 |

Continued on next page

Table S1 – continued from previous page

| Condition                | Method                     | Deletion | Bounded | Gap     | $n$  |
|--------------------------|----------------------------|----------|---------|---------|------|
| MovieLens full catalogue | Leave-one-out              | 1.0000   | 0.9580  | -0.0420 | 250  |
| MovieLens full catalogue | Monte Carlo Shapley        | 0.5742   | 0.6180  | 0.0438  | 250  |
| MovieLens full catalogue | Random control             | 0.0033   | 0.0039  | 0.0006  | 250  |
| MovieLens n_max=100      | Greedy sequential deletion | 0.3874   | 0.3807  | -0.0067 | 250  |
| MovieLens n_max=100      | LIME                       | 0.9280   | 0.9183  | -0.0097 | 250  |
| MovieLens n_max=100      | Leave-one-out              | 1.0000   | 0.9872  | -0.0128 | 250  |
| MovieLens n_max=100      | Monte Carlo Shapley        | 0.8018   | 0.8082  | 0.0064  | 250  |
| MovieLens n_max=100      | Random control             | 0.0010   | 0.0015  | 0.0005  | 250  |
| MovieLens n_max=50       | Greedy sequential deletion | 0.3694   | 0.3636  | -0.0058 | 250  |
| MovieLens n_max=50       | LIME                       | 0.9448   | 0.9344  | -0.0105 | 250  |
| MovieLens n_max=50       | Leave-one-out              | 1.0000   | 0.9867  | -0.0133 | 250  |
| MovieLens n_max=50       | Monte Carlo Shapley        | 0.7997   | 0.8079  | 0.0082  | 250  |
| MovieLens n_max=50       | Random control             | 0.0010   | 0.0013  | 0.0003  | 250  |
| MovieLens rho=0.25       | Greedy sequential deletion | 0.3780   | 0.3766  | -0.0014 | 250  |
| MovieLens rho=0.25       | LIME                       | 0.9551   | 0.9498  | -0.0053 | 250  |
| MovieLens rho=0.25       | Leave-one-out              | 1.0000   | 0.9922  | -0.0078 | 250  |
| MovieLens rho=0.25       | Monte Carlo Shapley        | 0.7596   | 0.7664  | 0.0068  | 250  |
| MovieLens rho=0.25       | Random control             | -0.0038  | -0.0050 | -0.0012 | 250  |
| MovieLens sampled        | Greedy sequential deletion | 0.3205   | 0.3183  | -0.0023 | 1000 |
| MovieLens sampled        | LIME                       | 0.9509   | 0.9334  | -0.0176 | 1000 |
| MovieLens sampled        | Leave-one-out              | 1.0000   | 0.9780  | -0.0220 | 1000 |

Continued on next page

Table S1 – continued from previous page

| Condition         | Method  |       | Deletion | Bounded | Gap    | $n$  |
|-------------------|---------|-------|----------|---------|--------|------|
| MovieLens sampled | Monte   | Carlo | 0.7617   | 0.7786  | 0.0169 | 1000 |
|                   | Shapley |       |          |         |        |      |
| MovieLens sampled | Random  | con-  | -0.0010  | 0.0003  | 0.0014 | 1000 |
|                   | trol    |       |          |         |        |      |

## S2 Joint outcomes (Table S2)

**Table S2:** Complete joint-intervention outcome matrix. The metric name identifies the utility and whether the quantity is an effect, success, abstention, or regret.

| Condition | Dataset   | Method                     | Metric                    | Mean    | $n$ |
|-----------|-----------|----------------------------|---------------------------|---------|-----|
| budget1   | MovieLens | Greedy sequential deletion | abstention                | 0.0000  | 250 |
| budget1   | MovieLens | Greedy sequential deletion | intervention success ndcg | 0.0640  | 250 |
| budget1   | MovieLens | Greedy sequential deletion | joint effect ndcg         | -0.0016 | 250 |
| budget1   | MovieLens | Greedy sequential deletion | normalized regret ndcg    | 0.9437  | 60  |
| budget1   | MovieLens | LIME                       | abstention                | 0.0040  | 250 |
| budget1   | MovieLens | LIME                       | intervention success ndcg | 0.1816  | 250 |
| budget1   | MovieLens | LIME                       | joint effect ndcg         | 0.0260  | 250 |
| budget1   | MovieLens | LIME                       | normalized regret ndcg    | 0.2966  | 60  |
| budget1   | MovieLens | Leave-one-out              | abstention                | 0.0040  | 250 |
| budget1   | MovieLens | Leave-one-out              | intervention success ndcg | 0.1880  | 250 |
| budget1   | MovieLens | Leave-one-out              | joint effect ndcg         | 0.0268  | 250 |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset   | Method                     | Metric                    | Mean   | $n$ |
|-----------|-----------|----------------------------|---------------------------|--------|-----|
| budget1   | MovieLens | Leave-one-out              | normalized regret<br>ndcg | 0.2601 | 60  |
| budget1   | MovieLens | Monte Carlo Shapley        | abstention                | 0.0000 | 250 |
| budget1   | MovieLens | Monte Carlo Shapley        | intervention success ndcg | 0.1944 | 250 |
| budget1   | MovieLens | Monte Carlo Shapley        | joint effect ndcg         | 0.0264 | 250 |
| budget1   | MovieLens | Monte Carlo Shapley        | normalized regret<br>ndcg | 0.2738 | 60  |
| budget1   | MovieLens | Random control             | abstention                | 0.0008 | 250 |
| budget1   | MovieLens | Random control             | intervention success ndcg | 0.0456 | 250 |
| budget1   | MovieLens | Random control             | joint effect ndcg         | 0.0009 | 250 |
| budget1   | MovieLens | Random control             | normalized regret<br>ndcg | 0.9598 | 60  |
| budget3   | MovieLens | Greedy sequential deletion | abstention                | 0.0000 | 250 |
| budget3   | MovieLens | Greedy sequential deletion | intervention success ndcg | 0.1520 | 250 |
| budget3   | MovieLens | Greedy sequential deletion | joint effect ndcg         | 0.0097 | 250 |

Continued on next page

Table S2 – continued from previous page

| Condition     | Dataset   | Method                     | Metric            | Mean   | $n$ |
|---------------|-----------|----------------------------|-------------------|--------|-----|
| budget3       | MovieLens | LIME                       | abstention        | 0.0040 | 250 |
| budget3       | MovieLens | LIME                       | intervention      | 0.3400 | 250 |
|               |           |                            | success ndcg      |        |     |
| budget3       | MovieLens | LIME                       | joint effect ndcg | 0.0620 | 250 |
| budget3       | MovieLens | Leave-one-out              | abstention        | 0.0040 | 250 |
|               |           |                            | intervention      | 0.3400 | 250 |
|               |           |                            | success ndcg      |        |     |
| budget3       | MovieLens | Leave-one-out              | joint effect ndcg | 0.0615 | 250 |
|               |           |                            | abstention        | 0.0000 | 250 |
| budget3       | MovieLens | Monte Carlo Shapley        | intervention      | 0.3400 | 250 |
|               |           |                            | success ndcg      |        |     |
| budget3       | MovieLens | Monte Carlo Shapley        | joint effect ndcg | 0.0622 | 250 |
|               |           |                            | abstention        | 0.0008 | 250 |
| budget3       | MovieLens | Random control             | intervention      | 0.0952 | 250 |
|               |           |                            | success ndcg      |        |     |
| budget3       | MovieLens | Random control             | joint effect ndcg | 0.0019 | 250 |
|               |           |                            | abstention        | 0.0000 | 250 |
| candidates100 | MovieLens | Greedy sequential deletion |                   |        |     |

Continued on next page

Table S2 – continued from previous page

| Condition     | Dataset   | Method                     | Metric                    | Mean    | $n$ |
|---------------|-----------|----------------------------|---------------------------|---------|-----|
| candidates100 | MovieLens | Greedy sequential deletion | intervention success ndcg | 0.0800  | 250 |
| candidates100 | MovieLens | Greedy sequential deletion | joint effect ndcg         | -0.0030 | 250 |
| candidates100 | MovieLens | Greedy sequential deletion | normalized regret ndcg    | 1.1995  | 94  |
| candidates100 | MovieLens | LIME                       | abstention                | 0.0000  | 250 |
| candidates100 | MovieLens | LIME                       | intervention success ndcg | 0.3344  | 250 |
| candidates100 | MovieLens | LIME                       | joint effect ndcg         | 0.0546  | 250 |
| candidates100 | MovieLens | LIME                       | normalized regret ndcg    | 0.1566  | 94  |
| candidates100 | MovieLens | Leave-one-out              | abstention                | 0.0000  | 250 |
| candidates100 | MovieLens | Leave-one-out              | intervention success ndcg | 0.3400  | 250 |
| candidates100 | MovieLens | Leave-one-out              | joint effect ndcg         | 0.0539  | 250 |
| candidates100 | MovieLens | Leave-one-out              | normalized regret ndcg    | 0.1373  | 94  |
| candidates100 | MovieLens | Monte Carlo Shapley        | abstention                | 0.0000  | 250 |
| candidates100 | MovieLens | Monte Carlo Shapley        | intervention success ndcg | 0.3320  | 250 |

Continued on next page



Table S2 – continued from previous page

| Condition     | Dataset   | Method                     | Metric                    | Mean   | <i>n</i> |
|---------------|-----------|----------------------------|---------------------------|--------|----------|
| candidates100 | MovieLens | Monte Carlo Shapley        | joint effect ndcg         | 0.0547 | 250      |
| candidates100 | MovieLens | Monte Carlo Shapley        | normalized regret ndcg    | 0.1654 | 94       |
| candidates100 | MovieLens | Random control             | abstention                | 0.0008 | 250      |
| candidates100 | MovieLens | Random control             | intervention success ndcg | 0.0960 | 250      |
| candidates100 | MovieLens | Random control             | joint effect ndcg         | 0.0057 | 250      |
| candidates100 | MovieLens | Random control             | normalized regret ndcg    | 1.1430 | 94       |
| candidates500 | MovieLens | Greedy sequential deletion | abstention                | 0.0000 | 250      |
| candidates500 | MovieLens | Greedy sequential deletion | intervention success ndcg | 0.1000 | 250      |
| candidates500 | MovieLens | Greedy sequential deletion | joint effect ndcg         | 0.0065 | 250      |
| candidates500 | MovieLens | Greedy sequential deletion | normalized regret ndcg    | 0.9937 | 60       |
| candidates500 | MovieLens | LIME                       | abstention                | 0.0000 | 250      |
| candidates500 | MovieLens | LIME                       | intervention success ndcg | 0.2120 | 250      |

Continued on next page

Table S2 – continued from previous page

| Condition     | Dataset   | Method                 | Metric                       | Mean   | $n$ |
|---------------|-----------|------------------------|------------------------------|--------|-----|
| candidates500 | MovieLens | LIME                   | joint effect ndcg            | 0.0297 | 250 |
| candidates500 | MovieLens | LIME                   | normalized regret<br>ndcg    | 0.1702 | 60  |
| candidates500 | MovieLens | Leave-one-out          | abstention                   | 0.0000 | 250 |
| candidates500 | MovieLens | Leave-one-out          | intervention<br>success ndcg | 0.2120 | 250 |
| candidates500 | MovieLens | Leave-one-out          | joint effect ndcg            | 0.0287 | 250 |
| candidates500 | MovieLens | Leave-one-out          | normalized regret<br>ndcg    | 0.1821 | 60  |
| candidates500 | MovieLens | Monte Carlo<br>Shapley | abstention                   | 0.0000 | 250 |
| candidates500 | MovieLens | Monte Carlo<br>Shapley | intervention<br>success ndcg | 0.1952 | 250 |
| candidates500 | MovieLens | Monte Carlo<br>Shapley | joint effect ndcg            | 0.0250 | 250 |
| candidates500 | MovieLens | Monte Carlo<br>Shapley | normalized regret<br>ndcg    | 0.2840 | 60  |
| candidates500 | MovieLens | Random control         | abstention                   | 0.0008 | 250 |
| candidates500 | MovieLens | Random control         | intervention<br>success ndcg | 0.0568 | 250 |
| candidates500 | MovieLens | Random control         | joint effect ndcg            | 0.0012 | 250 |
| candidates500 | MovieLens | Random control         | normalized regret<br>ndcg    | 1.4627 | 60  |

Continued on next page

Table S2 – continued from previous page

| Condition      | Dataset | Method                     | Metric                    | Mean   | $n$ |
|----------------|---------|----------------------------|---------------------------|--------|-----|
| full_catalogue | Amazon  | Greedy sequential deletion | abstention                | 0.0000 | 250 |
| full_catalogue | Amazon  | Greedy sequential deletion | intervention success ndcg | 0.0120 | 250 |
| full_catalogue | Amazon  | Greedy sequential deletion | joint effect ndcg         | 0.0025 | 250 |
| full_catalogue | Amazon  | Greedy sequential deletion | normalized regret ndcg    | 0.4194 | 5   |
| full_catalogue | Amazon  | LIME                       | abstention                | 0.4608 | 250 |
| full_catalogue | Amazon  | LIME                       | intervention success ndcg | 0.0120 | 250 |
| full_catalogue | Amazon  | LIME                       | joint effect ndcg         | 0.0026 | 250 |
| full_catalogue | Amazon  | LIME                       | normalized regret ndcg    | 0.4000 | 5   |
| full_catalogue | Amazon  | Leave-one-out              | abstention                | 0.4560 | 250 |
| full_catalogue | Amazon  | Leave-one-out              | intervention success ndcg | 0.0120 | 250 |
| full_catalogue | Amazon  | Leave-one-out              | joint effect ndcg         | 0.0026 | 250 |
| full_catalogue | Amazon  | Leave-one-out              | normalized regret ndcg    | 0.4000 | 5   |

Continued on next page

Table S2 – continued from previous page

| Condition      | Dataset   | Method                     | Metric            | Mean   | $n$ |
|----------------|-----------|----------------------------|-------------------|--------|-----|
| full_catalogue | Amazon    | Monte Carlo                | abstention        | 0.0000 | 250 |
|                |           | Shapley                    |                   |        |     |
| full_catalogue | Amazon    | Monte Carlo                | intervention      | 0.0152 | 250 |
|                |           | Shapley                    | success ndcg      |        |     |
| full_catalogue | Amazon    | Monte Carlo                | joint effect ndcg | 0.0035 | 250 |
|                |           | Shapley                    |                   |        |     |
| full_catalogue | Amazon    | Monte Carlo                | normalized regret | 0.2555 | 5   |
|                |           | Shapley                    | ndcg              |        |     |
| full_catalogue | Amazon    | Random control             | abstention        | 0.0528 | 250 |
| full_catalogue | Amazon    | Random control             | intervention      | 0.0048 | 250 |
|                |           |                            | success ndcg      |        |     |
| full_catalogue | Amazon    | Random control             | joint effect ndcg | 0.0005 | 250 |
|                |           |                            |                   |        |     |
| full_catalogue | Amazon    | Random control             | normalized regret | 0.7928 | 5   |
|                |           |                            | ndcg              |        |     |
| full_catalogue | MovieLens | Greedy sequential deletion | abstention        | 0.0000 | 250 |
|                |           |                            |                   |        |     |
| full_catalogue | MovieLens | Greedy sequential deletion | intervention      | 0.0440 | 250 |
|                |           |                            | success ndcg      |        |     |
| full_catalogue | MovieLens | Greedy sequential deletion | joint effect ndcg | 0.0046 | 250 |
|                |           |                            |                   |        |     |

Continued on next page

Table S2 – continued from previous page

| Condition      | Dataset   | Method                     | Metric                       | Mean   | $n$ |
|----------------|-----------|----------------------------|------------------------------|--------|-----|
| full_catalogue | MovieLens | Greedy sequential deletion | normalized regret<br>ndcg    | 0.6189 | 20  |
| full_catalogue | MovieLens | LIME                       | abstention                   | 0.0000 | 250 |
| full_catalogue | MovieLens | LIME                       | intervention<br>success ndcg | 0.0720 | 250 |
| full_catalogue | MovieLens | LIME                       | joint effect ndcg            | 0.0145 | 250 |
| full_catalogue | MovieLens | LIME                       | normalized regret<br>ndcg    | 0.2076 | 20  |
| full_catalogue | MovieLens | Leave-one-out              | abstention                   | 0.0000 | 250 |
| full_catalogue | MovieLens | Leave-one-out              | intervention<br>success ndcg | 0.0720 | 250 |
| full_catalogue | MovieLens | Leave-one-out              | joint effect ndcg            | 0.0143 | 250 |
| full_catalogue | MovieLens | Leave-one-out              | normalized regret<br>ndcg    | 0.2133 | 20  |
| full_catalogue | MovieLens | Monte Carlo Shapley        | abstention                   | 0.0000 | 250 |
| full_catalogue | MovieLens | Monte Carlo Shapley        | intervention<br>success ndcg | 0.0672 | 250 |
| full_catalogue | MovieLens | Monte Carlo Shapley        | joint effect ndcg            | 0.0115 | 250 |
| full_catalogue | MovieLens | Monte Carlo Shapley        | normalized regret<br>ndcg    | 0.2905 | 20  |
| full_catalogue | MovieLens | Random control             | abstention                   | 0.0008 | 250 |

Continued on next page

Table S2 – continued from previous page

| Condition      | Dataset   | Method                     | Metric                    | Mean    | $n$ |
|----------------|-----------|----------------------------|---------------------------|---------|-----|
| full_catalogue | MovieLens | Random control             | intervention success ndcg | 0.0208  | 250 |
| full_catalogue | MovieLens | Random control             | joint effect ndcg         | -0.0005 | 250 |
| full_catalogue | MovieLens | Random control             | normalized regret ndcg    | 1.2435  | 20  |
| nmax100        | MovieLens | Greedy sequential deletion | abstention                | 0.0000  | 250 |
| nmax100        | MovieLens | Greedy sequential deletion | intervention success ndcg | 0.0640  | 250 |
| nmax100        | MovieLens | Greedy sequential deletion | joint effect ndcg         | 0.0028  | 250 |
| nmax100        | MovieLens | Greedy sequential deletion | normalized regret ndcg    | 0.8313  | 55  |
| nmax100        | MovieLens | LIME                       | abstention                | 0.0040  | 250 |
| nmax100        | MovieLens | LIME                       | intervention success ndcg | 0.1720  | 250 |
| nmax100        | MovieLens | LIME                       | joint effect ndcg         | 0.0221  | 250 |
| nmax100        | MovieLens | LIME                       | normalized regret ndcg    | 0.2604  | 55  |
| nmax100        | MovieLens | Leave-one-out              | abstention                | 0.0040  | 250 |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset   | Method                     | Metric                    | Mean    | $n$ |
|-----------|-----------|----------------------------|---------------------------|---------|-----|
| nmax100   | MovieLens | Leave-one-out              | intervention success ndcg | 0.1760  | 250 |
| nmax100   | MovieLens | Leave-one-out              | joint effect ndcg         | 0.0219  | 250 |
| nmax100   | MovieLens | Leave-one-out              | normalized regret ndcg    | 0.2449  | 55  |
| nmax100   | MovieLens | Monte Carlo Shapley        | abstention                | 0.0000  | 250 |
| nmax100   | MovieLens | Monte Carlo Shapley        | intervention success ndcg | 0.1784  | 250 |
| nmax100   | MovieLens | Monte Carlo Shapley        | joint effect ndcg         | 0.0237  | 250 |
| nmax100   | MovieLens | Monte Carlo Shapley        | normalized regret ndcg    | 0.2283  | 55  |
| nmax100   | MovieLens | Random control             | abstention                | 0.0008  | 250 |
| nmax100   | MovieLens | Random control             | intervention success ndcg | 0.0376  | 250 |
| nmax100   | MovieLens | Random control             | joint effect ndcg         | -0.0025 | 250 |
| nmax100   | MovieLens | Random control             | normalized regret ndcg    | 1.1975  | 55  |
| nmax50    | MovieLens | Greedy sequential deletion | abstention                | 0.0000  | 250 |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset   | Method                     | Metric                    | Mean    | $n$ |
|-----------|-----------|----------------------------|---------------------------|---------|-----|
| nmax50    | MovieLens | Greedy sequential deletion | intervention success ndcg | 0.0680  | 250 |
| nmax50    | MovieLens | Greedy sequential deletion | joint effect ndcg         | -0.0050 | 250 |
| nmax50    | MovieLens | Greedy sequential deletion | normalized regret ndcg    | 2.0846  | 61  |
| nmax50    | MovieLens | LIME                       | abstention                | 0.0040  | 250 |
| nmax50    | MovieLens | LIME                       | intervention success ndcg | 0.2096  | 250 |
| nmax50    | MovieLens | LIME                       | joint effect ndcg         | 0.0298  | 250 |
| nmax50    | MovieLens | LIME                       | normalized regret ndcg    | 0.2090  | 61  |
| nmax50    | MovieLens | Leave-one-out              | abstention                | 0.0040  | 250 |
| nmax50    | MovieLens | Leave-one-out              | intervention success ndcg | 0.2120  | 250 |
| nmax50    | MovieLens | Leave-one-out              | joint effect ndcg         | 0.0295  | 250 |
| nmax50    | MovieLens | Leave-one-out              | normalized regret ndcg    | 0.2150  | 61  |
| nmax50    | MovieLens | Monte Carlo Shapley        | abstention                | 0.0000  | 250 |
| nmax50    | MovieLens | Monte Carlo Shapley        | intervention success ndcg | 0.2048  | 250 |

Continued on next page



Table S2 – continued from previous page

| Condition | Dataset   | Method                     | Metric                    | Mean    | <i>n</i> |
|-----------|-----------|----------------------------|---------------------------|---------|----------|
| nmax50    | MovieLens | Monte Carlo Shapley        | joint effect ndcg         | 0.0295  | 250      |
| nmax50    | MovieLens | Monte Carlo Shapley        | normalized regret ndcg    | 0.2271  | 61       |
| nmax50    | MovieLens | Random control             | abstention                | 0.0008  | 250      |
| nmax50    | MovieLens | Random control             | intervention success ndcg | 0.0512  | 250      |
| nmax50    | MovieLens | Random control             | joint effect ndcg         | -0.0024 | 250      |
| nmax50    | MovieLens | Random control             | normalized regret ndcg    | 1.5288  | 61       |
| primary   | Amazon    | Greedy sequential deletion | abstention                | 0.0070  | 1000     |
| primary   | Amazon    | Greedy sequential deletion | intervention success ndcg | 0.0900  | 1000     |
| primary   | Amazon    | Greedy sequential deletion | joint effect ndcg         | 0.0102  | 1000     |
| primary   | Amazon    | Greedy sequential deletion | normalized regret ndcg    | 0.7836  | 196      |
| primary   | Amazon    | LIME                       | abstention                | 0.0702  | 1000     |
| primary   | Amazon    | LIME                       | intervention success ndcg | 0.1712  | 1000     |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset | Method                 | Metric                       | Mean    | $n$  |
|-----------|---------|------------------------|------------------------------|---------|------|
| primary   | Amazon  | LIME                   | joint effect ndcg            | 0.0331  | 1000 |
| primary   | Amazon  | LIME                   | normalized regret<br>ndcg    | 0.1975  | 196  |
| primary   | Amazon  | Leave-one-out          | abstention                   | 0.0860  | 1000 |
| primary   | Amazon  | Leave-one-out          | intervention<br>success ndcg | 0.1670  | 1000 |
| primary   | Amazon  | Leave-one-out          | joint effect ndcg            | 0.0313  | 1000 |
| primary   | Amazon  | Leave-one-out          | normalized regret<br>ndcg    | 0.2325  | 196  |
| primary   | Amazon  | Monte Carlo<br>Shapley | abstention                   | 0.0102  | 1000 |
| primary   | Amazon  | Monte Carlo<br>Shapley | intervention<br>success ndcg | 0.1702  | 1000 |
| primary   | Amazon  | Monte Carlo<br>Shapley | joint effect ndcg            | 0.0333  | 1000 |
| primary   | Amazon  | Monte Carlo<br>Shapley | normalized regret<br>ndcg    | 0.2061  | 196  |
| primary   | Amazon  | Random control         | abstention                   | 0.0476  | 1000 |
| primary   | Amazon  | Random control         | intervention<br>success ndcg | 0.0660  | 1000 |
| primary   | Amazon  | Random control         | joint effect ndcg            | -0.0039 | 1000 |
| primary   | Amazon  | Random control         | normalized regret<br>ndcg    | 1.1971  | 196  |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset   | Method                     | Metric                    | Mean   | $n$  |
|-----------|-----------|----------------------------|---------------------------|--------|------|
| primary   | MovieLens | Greedy sequential deletion | abstention                | 0.0000 | 1000 |
| primary   | MovieLens | Greedy sequential deletion | intervention success ndcg | 0.1030 | 1000 |
| primary   | MovieLens | Greedy sequential deletion | joint effect ndcg         | 0.0058 | 1000 |
| primary   | MovieLens | Greedy sequential deletion | normalized regret ndcg    | 1.4710 | 339  |
| primary   | MovieLens | LIME                       | abstention                | 0.0000 | 1000 |
| primary   | MovieLens | LIME                       | intervention success ndcg | 0.2876 | 1000 |
| primary   | MovieLens | LIME                       | joint effect ndcg         | 0.0425 | 1000 |
| primary   | MovieLens | LIME                       | normalized regret ndcg    | 0.2254 | 339  |
| primary   | MovieLens | Leave-one-out              | abstention                | 0.0000 | 1000 |
| primary   | MovieLens | Leave-one-out              | intervention success ndcg | 0.2890 | 1000 |
| primary   | MovieLens | Leave-one-out              | joint effect ndcg         | 0.0431 | 1000 |
| primary   | MovieLens | Leave-one-out              | normalized regret ndcg    | 0.2174 | 339  |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset   | Method                           | Metric                       | Mean    | $n$  |
|-----------|-----------|----------------------------------|------------------------------|---------|------|
| primary   | MovieLens | Monte Carlo<br>Shapley           | abstention                   | 0.0000  | 1000 |
| primary   | MovieLens | Monte Carlo<br>Shapley           | intervention<br>success ndcg | 0.2742  | 1000 |
| primary   | MovieLens | Monte Carlo<br>Shapley           | joint effect ndcg            | 0.0403  | 1000 |
| primary   | MovieLens | Monte Carlo<br>Shapley           | normalized regret<br>ndcg    | 0.2675  | 339  |
| primary   | MovieLens | Random control                   | abstention                   | 0.0004  | 1000 |
| primary   | MovieLens | Random control                   | intervention<br>success ndcg | 0.0744  | 1000 |
| primary   | MovieLens | Random control                   | joint effect ndcg            | -0.0010 | 1000 |
| primary   | MovieLens | Random control                   | normalized regret<br>ndcg    | 1.4455  | 339  |
| rho025    | MovieLens | Greedy<br>sequential<br>deletion | abstention                   | 0.0000  | 250  |
| rho025    | MovieLens | Greedy<br>sequential<br>deletion | intervention<br>success ndcg | 0.1440  | 250  |
| rho025    | MovieLens | Greedy<br>sequential<br>deletion | joint effect ndcg            | 0.0060  | 250  |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset   | Method                     | Metric                       | Mean   | $n$ |
|-----------|-----------|----------------------------|------------------------------|--------|-----|
| rho025    | MovieLens | Greedy sequential deletion | normalized regret<br>ndcg    | 0.9987 | 98  |
| rho025    | MovieLens | LIME                       | abstention                   | 0.0040 | 250 |
| rho025    | MovieLens | LIME                       | intervention<br>success ndcg | 0.3576 | 250 |
| rho025    | MovieLens | LIME                       | joint effect ndcg            | 0.0651 | 250 |
| rho025    | MovieLens | LIME                       | normalized regret<br>ndcg    | 0.1565 | 98  |
| rho025    | MovieLens | Leave-one-out              | abstention                   | 0.0040 | 250 |
| rho025    | MovieLens | Leave-one-out              | intervention<br>success ndcg | 0.3600 | 250 |
| rho025    | MovieLens | Leave-one-out              | joint effect ndcg            | 0.0653 | 250 |
| rho025    | MovieLens | Leave-one-out              | normalized regret<br>ndcg    | 0.1440 | 98  |
| rho025    | MovieLens | Monte Carlo Shapley        | abstention                   | 0.0000 | 250 |
| rho025    | MovieLens | Monte Carlo Shapley        | intervention<br>success ndcg | 0.3584 | 250 |
| rho025    | MovieLens | Monte Carlo Shapley        | joint effect ndcg            | 0.0643 | 250 |
| rho025    | MovieLens | Monte Carlo Shapley        | normalized regret<br>ndcg    | 0.1691 | 98  |
| rho025    | MovieLens | Random control             | abstention                   | 0.0008 | 250 |

Continued on next page

Table S2 – continued from previous page

| Condition | Dataset   | Method         | Metric                    | Mean   | $n$ |
|-----------|-----------|----------------|---------------------------|--------|-----|
| rho025    | MovieLens | Random control | intervention success ndcg | 0.0944 | 250 |
| rho025    | MovieLens | Random control | joint effect ndcg         | 0.0002 | 250 |
| rho025    | MovieLens | Random control | normalized regret ndcg    | 1.2587 | 98  |

### S3 Paired comparisons (Tables S3a–S3c)

Table S3a: AIA and gap

**Table S3:** Supplementary Table S3a: deletion AIA, bounded AIA, and Actionability Gap comparisons.

| Dataset | Metric            | Left       | Right     | $\Delta$ | CI low  | CI high | Holm-adjusted $p$ | $d_z$   | $n$ |
|---------|-------------------|------------|-----------|----------|---------|---------|-------------------|---------|-----|
| Amazon  | Actionability Gap | greedy_cf  | random    | -0.0577  | -0.0750 | -0.0411 | 0.0010            | -0.2104 | 993 |
| Amazon  | Actionability Gap | lime       | greedy_cf | -0.0432  | -0.0586 | -0.0273 | 0.0010            | -0.1695 | 993 |
| Amazon  | Actionability Gap | lime       | loo       | 0.0457   | 0.0365  | 0.0555  | 0.0010            | 0.3006  | 993 |
| Amazon  | Actionability Gap | lime       | random    | -0.1009  | -0.1163 | -0.0860 | 0.0010            | -0.4122 | 993 |
| Amazon  | Actionability Gap | loo        | greedy_cf | -0.0888  | -0.1054 | -0.0730 | 0.0010            | -0.3373 | 993 |
| Amazon  | Actionability Gap | loo        | random    | -0.1465  | -0.1633 | -0.1304 | 0.0010            | -0.5540 | 993 |
| Amazon  | Actionability Gap | shapley_mc | greedy_cf | 0.1886   | 0.1594  | 0.2198  | 0.0010            | 0.3859  | 993 |
| Amazon  | Actionability Gap | shapley_mc | lime      | 0.2318   | 0.2021  | 0.2629  | 0.0010            | 0.4729  | 993 |
| Amazon  | Actionability Gap | shapley_mc | loo       | 0.2774   | 0.2452  | 0.3109  | 0.0010            | 0.5248  | 993 |
| Amazon  | Actionability Gap | shapley_mc | random    | 0.1309   | 0.1108  | 0.1515  | 0.0010            | 0.3920  | 993 |
| Amazon  | bounded AIA       | greedy_cf  | random    | 0.1413   | 0.1058  | 0.1766  | 0.0010            | 0.2492  | 993 |
| Amazon  | bounded AIA       | lime       | greedy_cf | 0.6961   | 0.6638  | 0.7301  | 0.0010            | 1.3169  | 993 |
| Amazon  | bounded AIA       | lime       | loo       | -0.0240  | -0.0317 | -0.0163 | 0.0010            | -0.1908 | 993 |
| Amazon  | bounded AIA       | lime       | random    | 0.8375   | 0.8166  | 0.8589  | 0.0010            | 2.4300  | 993 |
| Amazon  | bounded AIA       | loo        | greedy_cf | 0.7201   | 0.6878  | 0.7531  | 0.0010            | 1.3796  | 993 |

Continued on next page



Table S3a – continued from previous page

| Dataset   | Metric            | Left       | Right     | $\Delta$ | CI low  | CI high | Holm-adjusted $p$ | $d_z$   | $n$  |
|-----------|-------------------|------------|-----------|----------|---------|---------|-------------------|---------|------|
| Amazon    | bounded AIA       | loo        | random    | 0.8614   | 0.8394  | 0.8830  | 0.0010            | 2.4605  | 993  |
| Amazon    | bounded AIA       | shapley_mc | greedy_cf | 0.2831   | 0.2377  | 0.3278  | 0.0010            | 0.3878  | 993  |
| Amazon    | bounded AIA       | shapley_mc | lime      | -0.4130  | -0.4489 | -0.3782 | 0.0010            | -0.7258 | 993  |
| Amazon    | bounded AIA       | shapley_mc | loo       | -0.4370  | -0.4721 | -0.4019 | 0.0010            | -0.7711 | 993  |
| Amazon    | bounded AIA       | shapley_mc | random    | 0.4244   | 0.3874  | 0.4608  | 0.0010            | 0.7255  | 993  |
| Amazon    | deletion AIA      | greedy_cf  | random    | 0.1990   | 0.1642  | 0.2342  | 0.0010            | 0.3561  | 993  |
| Amazon    | deletion AIA      | lime       | greedy_cf | 0.7393   | 0.7080  | 0.7710  | 0.0010            | 1.4431  | 993  |
| Amazon    | deletion AIA      | lime       | loo       | -0.0696  | -0.0762 | -0.0635 | 0.0010            | -0.6841 | 993  |
| Amazon    | deletion AIA      | lime       | random    | 0.9383   | 0.9221  | 0.9546  | 0.0010            | 3.6155  | 993  |
| Amazon    | deletion AIA      | loo        | greedy_cf | 0.8089   | 0.7767  | 0.8406  | 0.0010            | 1.5799  | 993  |
| Amazon    | deletion AIA      | loo        | random    | 1.0080   | 0.9934  | 1.0226  | 0.0010            | 4.2381  | 993  |
| Amazon    | deletion AIA      | shapley_mc | greedy_cf | 0.0945   | 0.0451  | 0.1429  | 0.0010            | 0.1211  | 993  |
| Amazon    | deletion AIA      | shapley_mc | lime      | -0.6448  | -0.6836 | -0.6083 | 0.0010            | -1.0634 | 993  |
| Amazon    | deletion AIA      | shapley_mc | loo       | -0.7144  | -0.7523 | -0.6766 | 0.0010            | -1.1771 | 993  |
| Amazon    | deletion AIA      | shapley_mc | random    | 0.2935   | 0.2521  | 0.3340  | 0.0010            | 0.4488  | 993  |
| MovieLens | Actionability     | greedy_cf  | random    | -0.0036  | -0.0109 | 0.0018  | 0.3099            | -0.0351 | 1000 |
| MovieLens | Actionability Gap | lime       | greedy_cf | -0.0153  | -0.0194 | -0.0115 | 0.0010            | -0.2410 | 1000 |
| MovieLens | Actionability Gap | lime       | loo       | 0.0045   | 0.0031  | 0.0060  | 0.0010            | 0.1942  | 1000 |
| MovieLens | Actionability Gap | lime       | random    | -0.0190  | -0.0260 | -0.0142 | 0.0010            | -0.1958 | 1000 |
| MovieLens | Actionability Gap | loo        | greedy_cf | -0.0198  | -0.0237 | -0.0160 | 0.0010            | -0.3175 | 1000 |
| MovieLens | Actionability Gap | loo        | random    | -0.0234  | -0.0303 | -0.0186 | 0.0010            | -0.2427 | 1000 |

Continued on next page

Table S3a – continued from previous page

| Dataset   | Metric            | Left       | Right     | $\Delta$ | CI low  | CI high | Holm-adjusted $p$ | $d_z$   | $n$  |
|-----------|-------------------|------------|-----------|----------|---------|---------|-------------------|---------|------|
| MovieLens | Actionability Gap | shapley_mc | greedy_cf | 0.0191   | 0.0127  | 0.0273  | 0.0010            | 0.1596  | 1000 |
| MovieLens | Actionability Gap | shapley_mc | lime      | 0.0344   | 0.0281  | 0.0428  | 0.0010            | 0.2888  | 1000 |
| MovieLens | Actionability Gap | shapley_mc | loo       | 0.0389   | 0.0323  | 0.0475  | 0.0010            | 0.3194  | 1000 |
| MovieLens | Actionability Gap | shapley_mc | random    | 0.0155   | 0.0119  | 0.0193  | 0.0010            | 0.2611  | 1000 |
| MovieLens | bounded AIA       | greedy_cf  | random    | 0.3179   | 0.3000  | 0.3357  | 0.0010            | 1.0816  | 1000 |
| MovieLens | bounded AIA       | lime       | greedy_cf | 0.6151   | 0.5976  | 0.6327  | 0.0010            | 2.1833  | 1000 |
| MovieLens | bounded AIA       | lime       | loo       | -0.0446  | -0.0476 | -0.0416 | 0.0010            | -0.9264 | 1000 |
| MovieLens | bounded AIA       | lime       | random    | 0.9330   | 0.9241  | 0.9414  | 0.0010            | 6.5305  | 1000 |
| MovieLens | bounded AIA       | loo        | greedy_cf | 0.6597   | 0.6425  | 0.6769  | 0.0010            | 2.3881  | 1000 |
| MovieLens | bounded AIA       | loo        | random    | 0.9776   | 0.9696  | 0.9856  | 0.0010            | 7.6132  | 1000 |
| MovieLens | bounded AIA       | shapley_mc | greedy_cf | 0.4604   | 0.4402  | 0.4801  | 0.0010            | 1.4312  | 1000 |
| MovieLens | bounded AIA       | shapley_mc | lime      | -0.1547  | -0.1643 | -0.1452 | 0.0010            | -1.0100 | 1000 |
| MovieLens | bounded AIA       | shapley_mc | loo       | -0.1994  | -0.2095 | -0.1896 | 0.0010            | -1.2542 | 1000 |
| MovieLens | bounded AIA       | shapley_mc | random    | 0.7783   | 0.7662  | 0.7897  | 0.0010            | 4.0791  | 1000 |
| MovieLens | deletion AIA      | greedy_cf  | random    | 0.3216   | 0.3031  | 0.3405  | 0.0010            | 1.0794  | 1000 |
| MovieLens | deletion AIA      | lime       | greedy_cf | 0.6304   | 0.6136  | 0.6475  | 0.0010            | 2.2535  | 1000 |
| MovieLens | deletion AIA      | lime       | loo       | -0.0491  | -0.0518 | -0.0464 | 0.0010            | -1.1099 | 1000 |
| MovieLens | deletion AIA      | lime       | random    | 0.9520   | 0.9446  | 0.9593  | 0.0010            | 8.0043  | 1000 |
| MovieLens | deletion AIA      | loo        | greedy_cf | 0.6795   | 0.6623  | 0.6964  | 0.0010            | 2.4705  | 1000 |
| MovieLens | deletion AIA      | loo        | random    | 1.0010   | 0.9943  | 1.0080  | 0.0010            | 8.9894  | 1000 |
| MovieLens | deletion AIA      | shapley_mc | greedy_cf | 0.4412   | 0.4200  | 0.4619  | 0.0010            | 1.2995  | 1000 |
| MovieLens | deletion AIA      | shapley_mc | lime      | -0.1892  | -0.2001 | -0.1787 | 0.0010            | -1.1174 | 1000 |
| MovieLens | deletion AIA      | shapley_mc | loo       | -0.2383  | -0.2494 | -0.2271 | 0.0010            | -1.3229 | 1000 |

Continued on next page

Table S3a – continued from previous page

| Dataset   | Metric       | Left       | Right  | $\Delta$ | CI low | CI high | Holm-adjusted $p$ | $d_z$  | $n$  |
|-----------|--------------|------------|--------|----------|--------|---------|-------------------|--------|------|
| MovieLens | deletion AIA | shapley_mc | random | 0.7628   | 0.7499 | 0.7753  | 0.0010            | 3.7061 | 1000 |

Table S3b: effects and success

**Table S4:** Supplementary Table S3b: joint NDCG effects and success comparisons.

| Dataset   | Metric        | Left       | Right     | $\Delta$ | CI low  | CI high | Holm-adjusted $p$ | $d_z$   | $n$  |
|-----------|---------------|------------|-----------|----------|---------|---------|-------------------|---------|------|
| Amazon    | Success       | greedy_cf  | random    | 0.0240   | 0.0106  | 0.0376  | 0.0036            | 0.1094  | 1000 |
| Amazon    | Success       | lime       | greedy_cf | 0.0812   | 0.0628  | 0.1000  | 0.0010            | 0.2684  | 1000 |
| Amazon    | Success       | lime       | loo       | 0.0042   | 0.0002  | 0.0092  | 0.2256            | 0.0572  | 1000 |
| Amazon    | Success       | lime       | random    | 0.1052   | 0.0888  | 0.1224  | 0.0010            | 0.3941  | 1000 |
| Amazon    | Success       | loo        | greedy_cf | 0.0770   | 0.0590  | 0.0960  | 0.0010            | 0.2550  | 1000 |
| Amazon    | Success       | loo        | random    | 0.1010   | 0.0848  | 0.1180  | 0.0010            | 0.3766  | 1000 |
| Amazon    | Success       | shapley_mc | greedy_cf | 0.0802   | 0.0622  | 0.0990  | 0.0010            | 0.2684  | 1000 |
| Amazon    | Success       | shapley_mc | lime      | -0.0010  | -0.0072 | 0.0054  | 0.9045            | -0.0099 | 1000 |
| Amazon    | Success       | shapley_mc | loo       | 0.0032   | -0.0046 | 0.0112  | 0.9045            | 0.0251  | 1000 |
| Amazon    | Success       | shapley_mc | random    | 0.1042   | 0.0880  | 0.1206  | 0.0010            | 0.3969  | 1000 |
| Amazon    | $\Delta$ NDCG | greedy_cf  | random    | 0.0140   | 0.0098  | 0.0184  | 0.0010            | 0.1976  | 1000 |
| Amazon    | $\Delta$ NDCG | lime       | greedy_cf | 0.0230   | 0.0175  | 0.0288  | 0.0010            | 0.2533  | 1000 |
| Amazon    | $\Delta$ NDCG | lime       | loo       | 0.0018   | 0.0006  | 0.0033  | 0.0192            | 0.0818  | 1000 |
| Amazon    | $\Delta$ NDCG | lime       | random    | 0.0370   | 0.0314  | 0.0429  | 0.0010            | 0.4000  | 1000 |
| Amazon    | $\Delta$ NDCG | loo        | greedy_cf | 0.0212   | 0.0158  | 0.0267  | 0.0010            | 0.2368  | 1000 |
| Amazon    | $\Delta$ NDCG | loo        | random    | 0.0352   | 0.0296  | 0.0410  | 0.0010            | 0.3838  | 1000 |
| Amazon    | $\Delta$ NDCG | shapley_mc | greedy_cf | 0.0231   | 0.0177  | 0.0288  | 0.0010            | 0.2540  | 1000 |
| Amazon    | $\Delta$ NDCG | shapley_mc | lime      | 0.0001   | -0.0020 | 0.0023  | 0.8983            | 0.0039  | 1000 |
| Amazon    | $\Delta$ NDCG | shapley_mc | loo       | 0.0019   | -0.0006 | 0.0045  | 0.2598            | 0.0474  | 1000 |
| Amazon    | $\Delta$ NDCG | shapley_mc | random    | 0.0371   | 0.0315  | 0.0429  | 0.0010            | 0.3982  | 1000 |
| MovieLens | Success       | greedy_cf  | random    | 0.0286   | 0.0118  | 0.0460  | 0.0028            | 0.1025  | 1000 |
| MovieLens | Success       | lime       | greedy_cf | 0.1846   | 0.1606  | 0.2096  | 0.0010            | 0.4746  | 1000 |
| MovieLens | Success       | lime       | loo       | -0.0014  | -0.0064 | 0.0040  | 0.6355            | -0.0169 | 1000 |
| MovieLens | Success       | lime       | random    | 0.2132   | 0.1898  | 0.2362  | 0.0010            | 0.5748  | 1000 |
| MovieLens | Success       | loo        | greedy_cf | 0.1860   | 0.1620  | 0.2100  | 0.0010            | 0.4747  | 1000 |

Continued on next page

Table S3b – continued from previous page

| Dataset   | Metric        | Left       | Right     | $\Delta$ | CI low  | CI high | Holm-adjusted $p$ | $d_z$   | $n$  |
|-----------|---------------|------------|-----------|----------|---------|---------|-------------------|---------|------|
| MovieLens | Success       | loo        | random    | 0.2146   | 0.1916  | 0.2378  | 0.0010            | 0.5705  | 1000 |
| MovieLens | Success       | shapley_mc | greedy_cf | 0.1712   | 0.1480  | 0.1950  | 0.0010            | 0.4550  | 1000 |
| MovieLens | Success       | shapley_mc | lime      | -0.0134  | -0.0220 | -0.0050 | 0.0066            | -0.0974 | 1000 |
| MovieLens | Success       | shapley_mc | loo       | -0.0148  | -0.0248 | -0.0050 | 0.0066            | -0.0923 | 1000 |
| MovieLens | Success       | shapley_mc | random    | 0.1998   | 0.1774  | 0.2218  | 0.0010            | 0.5540  | 1000 |
| MovieLens | $\Delta$ NDCG | greedy_cf  | random    | 0.0068   | 0.0024  | 0.0113  | 0.0092            | 0.0959  | 1000 |
| MovieLens | $\Delta$ NDCG | lime       | greedy_cf | 0.0367   | 0.0311  | 0.0425  | 0.0010            | 0.3887  | 1000 |
| MovieLens | $\Delta$ NDCG | lime       | loo       | -0.0006  | -0.0019 | 0.0006  | 0.3632            | -0.0302 | 1000 |
| MovieLens | $\Delta$ NDCG | lime       | random    | 0.0435   | 0.0382  | 0.0491  | 0.0010            | 0.4964  | 1000 |
| MovieLens | $\Delta$ NDCG | loo        | greedy_cf | 0.0373   | 0.0315  | 0.0433  | 0.0010            | 0.3904  | 1000 |
| MovieLens | $\Delta$ NDCG | loo        | random    | 0.0441   | 0.0386  | 0.0498  | 0.0010            | 0.4918  | 1000 |
| MovieLens | $\Delta$ NDCG | shapley_mc | greedy_cf | 0.0344   | 0.0287  | 0.0401  | 0.0010            | 0.3742  | 1000 |
| MovieLens | $\Delta$ NDCG | shapley_mc | lime      | -0.0023  | -0.0040 | -0.0007 | 0.0144            | -0.0854 | 1000 |
| MovieLens | $\Delta$ NDCG | shapley_mc | loo       | -0.0029  | -0.0051 | -0.0008 | 0.0144            | -0.0847 | 1000 |
| MovieLens | $\Delta$ NDCG | shapley_mc | random    | 0.0413   | 0.0360  | 0.0467  | 0.0010            | 0.4840  | 1000 |

Table S3c: normalized regret

**Table S5:** Supplementary Table S3c: conditional normalized NDCG-regret comparisons.

| Dataset   | Metric  | Left       | Right     | $\Delta$ | CI low  | CI high | Holm-adjusted $p$ | $d_z$   | $n$ |
|-----------|---------|------------|-----------|----------|---------|---------|-------------------|---------|-----|
| Amazon    | NRegret | greedy_cf  | random    | -0.4135  | -0.6029 | -0.2365 | 0.0010            | -0.3143 | 196 |
| Amazon    | NRegret | lime       | greedy_cf | -0.5862  | -0.7598 | -0.4379 | 0.0010            | -0.5010 | 196 |
| Amazon    | NRegret | lime       | loo       | -0.0350  | -0.0628 | -0.0105 | 0.0267            | -0.1853 | 196 |
| Amazon    | NRegret | lime       | random    | -0.9997  | -1.1707 | -0.8503 | 0.0010            | -0.8633 | 196 |
| Amazon    | NRegret | loo        | greedy_cf | -0.5512  | -0.7268 | -0.4014 | 0.0010            | -0.4678 | 196 |
| Amazon    | NRegret | loo        | random    | -0.9647  | -1.1438 | -0.8120 | 0.0010            | -0.8142 | 196 |
| Amazon    | NRegret | shapley_mc | greedy_cf | -0.5776  | -0.7595 | -0.4286 | 0.0010            | -0.4898 | 196 |
| Amazon    | NRegret | shapley_mc | lime      | 0.0086   | -0.0260 | 0.0431  | 0.6304            | 0.0348  | 196 |
| Amazon    | NRegret | shapley_mc | loo       | -0.0264  | -0.0689 | 0.0154  | 0.4590            | -0.0861 | 196 |
| Amazon    | NRegret | shapley_mc | random    | -0.9910  | -1.1595 | -0.8404 | 0.0010            | -0.8664 | 196 |
| MovieLens | NRegret | greedy_cf  | random    | 0.0255   | -0.2721 | 0.3421  | 0.8724            | 0.0089  | 339 |
| MovieLens | NRegret | lime       | greedy_cf | -1.2456  | -1.6561 | -0.8910 | 0.0010            | -0.3453 | 339 |
| MovieLens | NRegret | lime       | loo       | 0.0081   | -0.0112 | 0.0258  | 0.7967            | 0.0471  | 339 |
| MovieLens | NRegret | lime       | random    | -1.2201  | -1.4736 | -0.9952 | 0.0010            | -0.5432 | 339 |
| MovieLens | NRegret | loo        | greedy_cf | -1.2537  | -1.6573 | -0.8955 | 0.0010            | -0.3461 | 339 |
| MovieLens | NRegret | loo        | random    | -1.2281  | -1.4810 | -1.0027 | 0.0010            | -0.5427 | 339 |
| MovieLens | NRegret | shapley_mc | greedy_cf | -1.2035  | -1.6265 | -0.8482 | 0.0010            | -0.3337 | 339 |
| MovieLens | NRegret | shapley_mc | lime      | 0.0421   | 0.0173  | 0.0677  | 0.0064            | 0.1736  | 339 |
| MovieLens | NRegret | shapley_mc | loo       | 0.0501   | 0.0190  | 0.0818  | 0.0064            | 0.1694  | 339 |
| MovieLens | NRegret | shapley_mc | random    | -1.1780  | -1.4312 | -0.9610 | 0.0010            | -0.5221 | 339 |



## S4 Convergence matrix (Table S4)

**Table S6:** Supplementary convergence matrix.  $M_{\text{pair}}$  counts base permutations and  $T$  counts forward/reverse evaluated orders. Rank-valid fraction is the fraction of users with finite rank comparisons; threshold coverage is the fraction of those users satisfying both rank and action-overlap thresholds.

| Dataset   | Model   | Utility       | $M_{\text{pair}}$ | $T$  | Rank   | Jaccard | Rank-valid | Coverage | Selected?   |
|-----------|---------|---------------|-------------------|------|--------|---------|------------|----------|-------------|
| Amazon    | itemknn | target_margin | 25                | 50   | 0.9348 | 0.8707  | 100.0%     | 43.0%    | No          |
| Amazon    | itemknn | target_margin | 50                | 100  | 0.9532 | 0.9090  | 100.0%     | 60.0%    | Yes         |
| Amazon    | itemknn | target_margin | 100               | 200  | 0.9690 | 0.9450  | 100.0%     | 70.0%    | No          |
| Amazon    | itemknn | target_margin | 250               | 500  | 0.9739 | 0.9467  | 100.0%     | 78.0%    | No          |
| Amazon    | itemknn | target_margin | 500               | 1000 | 0.9797 | 0.9617  | 100.0%     | 83.0%    | No          |
| Amazon    | itemknn | target_margin | 1000              | 2000 | 0.9821 | 0.9653  | 100.0%     | 85.0%    | No          |
| Amazon    | profile | target_margin | 25                | 50   | 0.9023 | 0.7913  | 100.0%     | 23.0%    | No          |
| Amazon    | profile | target_margin | 50                | 100  | 0.9297 | 0.8563  | 100.0%     | 39.0%    | No          |
| Amazon    | profile | target_margin | 100               | 200  | 0.9670 | 0.9110  | 100.0%     | 66.0%    | Yes         |
| Amazon    | profile | target_margin | 250               | 500  | 0.9742 | 0.9297  | 100.0%     | 74.0%    | No          |
| Amazon    | profile | target_margin | 500               | 1000 | 0.9841 | 0.9467  | 100.0%     | 78.0%    | No          |
| Amazon    | profile | target_margin | 1000              | 2000 | 0.9888 | 0.9683  | 100.0%     | 89.0%    | No          |
| MovieLens | itemknn | ndcg          | 25                | 50   | 0.8257 | 0.6087  | 92.0%      | 22.8%    | No          |
| MovieLens | itemknn | ndcg          | 50                | 100  | 0.8575 | 0.6113  | 92.0%      | 27.2%    | No          |
| MovieLens | itemknn | ndcg          | 100               | 200  | 0.8847 | 0.6897  | 92.0%      | 39.1%    | No          |
| MovieLens | itemknn | ndcg          | 250               | 500  | 0.9167 | 0.7373  | 92.0%      | 52.2%    | No          |
| MovieLens | itemknn | ndcg          | 500               | 1000 | 0.9306 | 0.7740  | 92.0%      | 60.9%    | No          |
| MovieLens | itemknn | ndcg          | 1000              | 2000 | 0.9409 | 0.7813  | 92.0%      | 62.0%    | Unconverged |
|           |         |               |                   |      |        |         |            |          | max         |
| MovieLens | itemknn | target_margin | 25                | 50   | 0.9348 | 0.6467  | 100.0%     | 13.0%    | No          |
| MovieLens | itemknn | target_margin | 50                | 100  | 0.9584 | 0.7280  | 100.0%     | 33.0%    | No          |
| MovieLens | itemknn | target_margin | 100               | 200  | 0.9745 | 0.7880  | 100.0%     | 52.0%    | No          |

Continued on next page

Table S6 – continued from previous page

| Dataset   | Model   | Utility       | $M_{\text{pair}}$ | $T$  | Rank   | Jaccard | Rank-valid | Coverage | Selected? |
|-----------|---------|---------------|-------------------|------|--------|---------|------------|----------|-----------|
| MovieLens | itemknn | target_margin | 250               | 500  | 0.9853 | 0.8433  | 100.0%     | 69.0%    | Yes       |
| MovieLens | itemknn | target_margin | 500               | 1000 | 0.9902 | 0.8767  | 100.0%     | 74.0%    | No        |
| MovieLens | itemknn | target_margin | 1000              | 2000 | 0.9936 | 0.9040  | 100.0%     | 81.0%    | No        |
| MovieLens | profile | target_margin | 25                | 50   | 0.9455 | 0.7787  | 100.0%     | 25.0%    | No        |
| MovieLens | profile | target_margin | 50                | 100  | 0.9670 | 0.8320  | 100.0%     | 56.0%    | Yes       |
| MovieLens | profile | target_margin | 100               | 200  | 0.9779 | 0.8773  | 100.0%     | 70.0%    | No        |
| MovieLens | profile | target_margin | 250               | 500  | 0.9873 | 0.9053  | 100.0%     | 82.0%    | No        |
| MovieLens | profile | target_margin | 500               | 1000 | 0.9912 | 0.9267  | 100.0%     | 86.0%    | No        |
| MovieLens | profile | target_margin | 1000              | 2000 | 0.9941 | 0.9373  | 100.0%     | 89.0%    | No        |

## S5 Executable contract (Table S5)

**Table S7:** Expanded executable contract used by every final run.

| Parameter             | Frozen value                                                                                                       |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Temporal split        | Last event test; penultimate event validation; ties sorted by timestamp and original record index.                 |
| Training histories    | Complete histories are used to fit ItemKNN similarities and profile embeddings.                                    |
| Scoring/player window | Most recent 20 training interactions; older training context is not added as fixed scoring context.                |
| Candidates            | Target plus 199 uniformly sampled unseen negatives; complete pre-test history including validation is excluded.    |
| Models                | ItemKNN with 200 top non-zero cosine neighbours; 64-dimensional profile aggregation robustness model.              |
| Seeds                 | Model/attribution 42–46; candidate 1729; user 2718; tie 31415.                                                     |
| Attribution           | Target-margin utility with top $L = 10$ competitors; $M_{\text{pair}} = 500$ and $T = 1000$ reverse-paired orders. |
| LIME                  | 512 masks; normalized Hamming kernel width .25; ridge coefficient 1.0.                                             |
| Intervention          | Simultaneous $w_p \leftarrow 0.5w_p$ ; deletion and $\rho = .25$ sensitivity conditions.                           |
| Decision              | No action, all singletons and all pairs; exact oracle at $B = 2$ .                                                 |
| Inference             | 10,000 user bootstrap draws; 10,000 paired permutations; 1,000 matched AIA-null draws.                             |